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Trigonometric Identities and Equations

Get one trig function equal to a number. Change the range first when the angle is $ax + b$, find every solution with CAST, then undo the change.

THE TOPIC ON ONE MAP

Trigonometric Identities and Equations

01 — Transformed angle

new range CAST undo $ax + b$

02 — The two identities

$\tan = \frac{\sin}{\cos}$ $\sin^2 + \cos^2 = 1$ factorise, never divide

03 — Models and graphs

max and min period first time after ...

THE TWO IDENTITIES

$$\tan \theta \equiv \frac{\sin \theta}{\cos \theta}$$

$$\sin^2 \theta + \cos^2 \theta \equiv 1$$

$$\sin \theta = k \cos \theta \Rightarrow \tan \theta = k \quad \cdot \quad \sin^2 \text{ with } \cos \Rightarrow 1 - \cos^2 \quad \cdot \quad \tan \times \cos = \sin$$

01 Solve $\sin(ax + b) = k$: New Range, Then CAST

Call the whole angle $X = ax + b$ and change the range for X first. Use the calculator for the principal value, find every other value in the new range with CAST (or the graph), then solve back for x .

WHEN YOU SEE → YOU DO

$$3 \sin(2\theta - 10^\circ) = 1 \text{ for } 0 \leq \theta < 180^\circ$$

$$\rightarrow X = 2\theta - 10: -10 \leq X < 350$$

$$\tan^2\left(2x + \frac{\pi}{4}\right) = 3$$

$$\rightarrow \tan = \pm\sqrt{3}: \text{both signs}$$

$$3 \sin(\dots) = 7 \cos(\dots)$$

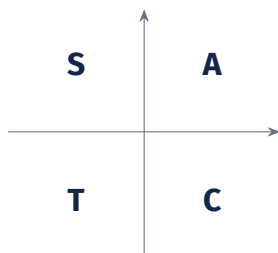
$$\rightarrow \text{divide by } \cos: \tan(\dots) = \frac{7}{3}$$

“give the answers in the form $k\pi$ ”

$$\rightarrow \text{exact values: } \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}} \dots$$

“in radians”

$$\rightarrow \text{calculator in radians; the range uses } \pi$$



$\sin$: α and $180^\circ - \alpha$

$\cos$: α and $360^\circ - \alpha$

$\tan$: α and $180^\circ + \alpha$

then add or subtract 360° to fill the range

WORKED MODEL • OCTOBER 2019

Solve $3 \sin(2\theta - 10^\circ) = 1$ for $0 \leq \theta < 180^\circ$.

1 $\sin X = \frac{1}{3}, -10 \leq X < 350: X = 19.47, 160.53$

new range, CAST

ANSWER $\theta = 14.7^\circ, 85.3^\circ$

IDEA 01

Practise: Solve $\sin(ax + b) = k$: New Range, Then CAST

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Transformed angle

P2 · TOPIC 10 4 marks

Solve $2 \cos(2x - 30^\circ) = 1$ for $0 \leq x < 360^\circ$.

Let $X = 2x - 30$: new range $-30 \leq X < 690$.

Q2 **DR ESLAM** Sine equals k cosine

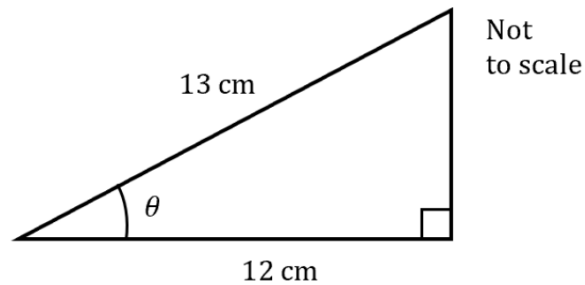
P2 · TOPIC 10 3 marks

Solve $\sin \theta = 3 \cos \theta$ for $0 \leq \theta < 360^\circ$.

SELECTED PRACTICE

Q3**PRACTICE** Practice · Easy**P2 · TOPIC 10** 5 marks**1 (a)**

(a) Work out the length of the missing side in the following right-angled triangle.

**(2 marks)****(b)**

(b) Using your answer from part (a) to help, write down the values of the following:

(i) $\sin \theta$

(ii) $\cos \theta$

(iii) $\tan \theta$

(3 marks)

Q4**PRACTICE**

Practice · Medium

P2 · TOPIC 10**6 marks**

- 6 A right-angled triangle has hypotenuse 8 cm. One of its other sides is 5 cm.

Find exact values for $\sin \theta$, $\cos \theta$ and $\tan \theta$, where θ is the smallest angle in the triangle.

(6 marks)

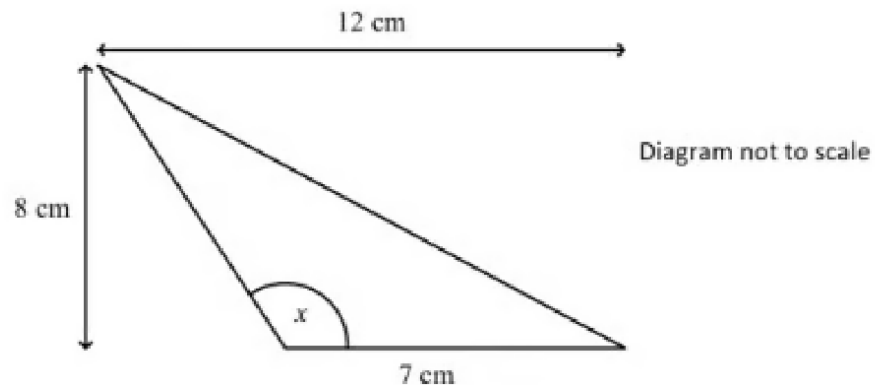
Q5**PRACTICE**

Practice · Very Hard

P2 · TOPIC 10

6 marks

6 For the triangle in the diagram, find exact values for $\sin x$, $\cos x$ and $\tan x$.



(6 marks)

PAST PAPERS

Q6 PAST PAPER Pearson IAL WMA12 · October 2019 · Q9

P2 · TOPIC 10 12 marks

9. Solutions based entirely on graphical or numerical methods are not acceptable in this question.

- (i) Solve, for $0 \leq \theta < 180^\circ$, the equation

$$3 \sin(2\theta - 10^\circ) = 1$$

giving your answers to one decimal place.

(4)

- (ii) The first three terms of an arithmetic sequence are

$$\sin \alpha, \frac{1}{\tan \alpha} \text{ and } 2 \sin \alpha$$

where α is a constant.

- (a) Show that $2 \cos \alpha = 3 \sin^2 \alpha$

(3)

Given that $\pi < \alpha < 2\pi$,

- (b) find, showing all working, the value of α to 3 decimal places.

(5)

(Total 12 marks)

Q7**PAST PAPER**

Pearson IAL WMA12 · October 2021 · Q10

P2 · TOPIC 10 10 marks

10. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

$$\tan^2\left(2x + \frac{\pi}{4}\right) = 3 \quad (5)$$

- (ii) Solve, for $0 < \theta < 360^\circ$

$$(2 \sin \theta - \cos \theta)^2 = 1$$

giving your answers, as appropriate, to one decimal place. (5)

Q8 **PAST PAPER** Pearson IAL WMA12 · January 2022 · Q7**P2 · TOPIC 10** **8 marks**

7. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $-90^\circ < x < 90^\circ$, the equation

$$3 \sin(2x - 15^\circ) = \cos(2x - 15^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) Solve, for $0 < \theta < 2\pi$, the equation

$$4 \sin^2 \theta + 8 \cos \theta = 3$$

giving your answers to 3 significant figures.

(4)

(Total 8 marks)

Q9**PAST PAPER**

Pearson IAL WMA12 · January 2023 · Q8

P2 · TOPIC 10 9 marks

8. In this question you must show all stages of your working.

Solutions based entirely on calculator technology are not acceptable.

- (i) Solve, for $-\frac{\pi}{2} < x < \pi$, the equation

$$5 \sin(3x + 0.1) + 2 = 0$$

giving your answers, **in radians**, to 2 decimal places.

(4)

- (ii) Solve, for $0 < \theta < 360^\circ$, the equation

$$2 \tan \theta \sin \theta = 5 + \cos \theta$$

giving your answers, **in degrees**, to one decimal place.

(5)

(Total for Question 8 is 9 marks)

Q10**PAST PAPER**

Pearson IAL WMA12 · June 2025 · Q8

P2 · TOPIC 10**8 marks****8:**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < \pi$, the equation

$$3 \tan \left(2x + \frac{\pi}{5} \right) = \sqrt{3}$$

giving the answers in radians in the form $k\pi$, where k is a rational constant to be found.

(3)

- (ii) Solve, for $0 \leq \theta < 360^\circ$, the equation

$$5 \sin \theta \tan \theta = \cos \theta + 4$$

giving your answers, in degrees, to one decimal place.

(5)

(Total for Question 8 is 8 marks)

Q11**PAST PAPER**

Pearson IAL WMA12 · January 2026 · Q8

P2 · TOPIC 10 10 marks**8.**

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

- (i) Solve, for $0 \leq \theta < 180$

$$4 \tan^2(2\theta - 30)^\circ + 1 = 49$$

giving the answers to one decimal place.

(5)

- (ii) Solve, for $0 \leq x < 2\pi$

$$2 \tan x \sin x + 3 = 0$$

giving the answers, in radians, in the form $k\pi$ where k is a rational constant.

(5)

(Total for Question 8 is 10 marks)

Q12**PAST PAPER**

Pearson IAL WMA12 · June 2026 · Q2

P2 · TOPIC 10**6 marks****2.**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < x \leq 2\pi$, the equation

$$1 + \sin^2 x + \cos x = 0 \quad (3)$$

- (ii) Solve, for $0 < \theta < 180^\circ$, the equation

$$\sin(\theta + 60^\circ) = 4 \cos(\theta + 60^\circ)$$

giving the answer to one decimal place.

(3)

(Total for Question 2 is 6 marks)

02 The Two Identities: Make One Trig Function

If the equation mixes $\sin$, $\cos$ and $\tan$, use the identities until only one function is left. Usually you get a quadratic (or cubic) in $\cos x$ or $\sin x$: factorise, and reject values outside $-1 \leq \text{value} \leq 1$.

WHEN YOU SEE → YOU DO

$\tan \theta$ **times something**

→ $\tan = \frac{\sin}{\cos}$, multiply through by $\cos$

$\sin^2 x$ **and** $\cos x$ **together**

→ $\sin^2 = 1 - \cos^2$: quadratic in $\cos$

“show that it can be written as
 $3 \sin^2 \theta + \sin \theta - 1 = 0$ ”

→ then “hence solve” with $2x$ in place of θ

$\sin x$ **on both sides**

→ bring to one side and factor — never divide by $\sin x$

$\cos x = 2$ **or** $\sin x = -5$

→ no solutions: reject

“given $\cos \theta = \frac{1}{3}$, **find** $\tan \theta$ **exactly”**

→ right triangle 1, 3, $\sqrt{8}$; sign from the quadrant

One function only

Quadratic

Factorise

CAST each root

WORKED MODEL • JUNE 2026

Solve $1 + \sin^2 x + \cos x = 0$ for $0 < x \leq 2\pi$.

1 $1 + (1 - \cos^2 x) + \cos x = 0 \Rightarrow \cos^2 x - \cos x - 2 = 0$

one function

2 $(\cos x - 2)(\cos x + 1) = 0$: $\cos x = -1$ only

reject $\cos x = 2$

ANSWER $x = \pi$

× **TRAP** Dividing both sides by $\sin x$ and losing the solutions of $\sin x = 0$.

IDEA 02

Practise: The Two Identities: Make One Trig Function

My questions first, then selected practice, then real past papers.

Q13 **DR ESLAM** One function only

P2 · TOPIC 10 4 marks

Solve $2 \sin^2 x = 3 \cos x$ for $0 \leq x < 360^\circ$.

$\sin^2 x = 1 - \cos^2 x$, then a quadratic in $\cos x$.

Q14 **DR ESLAM** Never divide by $\sin$

P2 · TOPIC 10 4 marks

Solve $\sin x \cos x = \frac{1}{2} \sin x$ for $0 \leq x \leq 180^\circ$.

SELECTED PRACTICE

Q15 **PRACTICE** Practice · Easy

P2 · TOPIC 10 2 marks

2 Show that

$$\frac{1 - \cos^2 x}{\tan^2 x} \equiv \cos^2 x$$

(2 marks)

Q16 **PRACTICE** Practice · Hard

P2 · TOPIC 10 5 marks

2 Solve the equation $2 \sin^2 \theta = \cos \theta + 1$ for $-180^\circ \leq \theta \leq 180^\circ$.

(5 marks)

Q17**PRACTICE**

Practice · Very Hard

P2 · TOPIC 10 12 marks**8 (a)** Find all the solutions to the equation

$2 \cos 2\theta = 4 \sin 2\theta \cos 2\theta$ in the interval $0 \leq \theta \leq 2\pi$, giving your answers in radians as multiples of π .

(6 marks)

(b) Find all the solutions to the equation $3\cos^2 4x + 13\cos 4x - 10 = 0$ in the interval $0 \leq x \leq \pi$, giving your answers in radians to three significant figures.

(6 marks)

PAST PAPERS

Q18 PAST PAPER Pearson IAL WMA12 · January 2020 · Q7

P2 · TOPIC 10 7 marks

7. (a) Show that the equation

$$8 \tan \theta = 3 \cos \theta$$

may be rewritten in the form

$$3 \sin^2 \theta + 8 \sin \theta - 3 = 0 \quad (3)$$

- (b) Hence solve, for $0 \leq x \leq 90^\circ$, the equation

$$8 \tan 2x = 3 \cos 2x$$

giving your answers to 2 decimal places. (4)

(Total 7 marks)

Q19**PAST PAPER**

Pearson IAL WMA12 · January 2021 · Q6

P2 · TOPIC 10**8 marks**

6. (a) Show that the equation

$$\frac{3 \sin \theta \cos \theta}{2 \sin \theta - 1} = 5 \tan \theta \quad \sin \theta \neq \frac{1}{2}$$

can be written in the form

$$3 \sin^3 \theta + 10 \sin^2 \theta - 8 \sin \theta = 0$$

(4)

- (b) Hence solve, for $-\frac{\pi}{4} < x < \frac{\pi}{4}$

$$\frac{3 \sin 2x \cos 2x}{2 \sin 2x - 1} = 5 \tan 2x$$

giving your answers to 3 decimal places where appropriate.

(4)

(Total 8 marks)

Q20**PAST PAPER**

Pearson IAL WMA12 · June 2021 · Q8

P2 · TOPIC 10**10 marks**

8. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < \theta < 360^\circ$, the equation

$$3 \sin(\theta + 30^\circ) = 7 \cos(\theta + 30^\circ)$$

giving your answers to one decimal place.

(4)

- (ii) (a) Show that the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

can be written in the form

$$\sin x (a \cos^2 x + b \cos x + c) = 0$$

where a , b and c are constants to be found.

- (b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$ the equation

$$3 \sin^3 x = 5 \sin x - 7 \sin x \cos x$$

(6)

(Total 10 marks)

Q21**PAST PAPER**

Pearson IAL WMA12 · June 2022 · Q5

P2 · TOPIC 10**6 marks**

5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Solve, for $-180^\circ < \theta \leq 180^\circ$, the equation

$$3 \tan(\theta + 43^\circ) = 2 \cos(\theta + 43^\circ)$$

(6)

(Total 6 marks)

Q22**PAST PAPER**

Pearson IAL WMA12 · October 2022 · Q5

P2 · TOPIC 10**8 marks**

5. **In this question you must show detailed reasoning.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$(3 \cos \theta - \tan \theta) \cos \theta = 2$$

can be written as

$$3 \sin^2 \theta + \sin \theta - 1 = 0 \quad (3)$$

(b) Hence solve for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$

$$(3 \cos 2x - \tan 2x) \cos 2x = 2 \quad (5)$$

(Total for Question 5 is 8 marks)

Q23**PAST PAPER**

Pearson IAL WMA12 · June 2023 · Q9

P2 · TOPIC 10 7 marks

9. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$3 \cos \theta (\tan \theta \sin \theta + 3) = 11 - 5 \cos \theta$$

may be written as

$$3 \cos^2 \theta - 14 \cos \theta + 8 = 0 \quad (3)$$

(b) Hence solve, for $0 < x < 360^\circ$

$$3 \cos 2x (\tan 2x \sin 2x + 3) = 11 - 5 \cos 2x$$

giving your answers to one decimal place. (4)

(Total for Question 9 is 7 marks)

Q24**PAST PAPER**

Pearson IAL WMA12 · October 2023 · Q3

P2 · TOPIC 10**7 marks**

3. **In this question you must show all stages of your working.**

Solutions relying entirely on calculator technology are not acceptable.

- (a) Solve, for $0 < \theta \leq 360^\circ$ the equation

$$2 \tan \theta + 3 \sin \theta = 0$$

giving your answers, as appropriate, to one decimal place.

(5)

- (b) Hence, or otherwise, find the smallest positive solution of

$$2 \tan(2x + 40^\circ) + 3 \sin(2x + 40^\circ) = 0$$

giving your answer to one decimal place.

(2)

(Total for Question 3 is 7 marks)

Q25**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q7

P2 · TOPIC 10 7 marks

7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < 2\pi$, the equation

$$3 \sin x \tan x = 11 + \cos x$$

giving the answers in radians to 3 decimal places.

(5)

- (ii) Given that

- $0 < \theta < 90^\circ$
- $\cos \theta = \frac{1}{3}$

find, in simplest form, the exact value of $\tan \theta$

(2)

(Total for Question 7 is 7 marks)

Q26**PAST PAPER**

Pearson IAL WMA12 · October 2024 · Q9

P2 · TOPIC 10**7 marks**

9. **In this question you must show detailed reasoning.**

Solutions relying entirely on calculator technology are not acceptable.

(a) Show that the equation

$$2 \tan \theta = 3 \cos \theta$$

can be written as

$$3 \sin^2 \theta + 2 \sin \theta - 3 = 0$$

(3)

(b) Hence solve, for $-\pi < x < \pi$, the equation

$$2 \tan \left(2x + \frac{\pi}{3} \right) = 3 \cos \left(2x + \frac{\pi}{3} \right)$$

giving your answers to 3 significant figures.

(4)

(Total for Question 9 is 7 marks)

Q27**PAST PAPER**

Pearson IAL WMA12 · January 2025 · Q10

P2 · TOPIC 10**8 marks****10.**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(a) Show that

$$\cos \theta \left(3 \tan \theta + \frac{2}{\tan \theta} \right) \equiv \sin \theta + \frac{2}{\sin \theta} \quad \theta \neq \frac{n\pi}{2} \quad (4)$$

(b) Hence solve, for $0 < x < 2\pi$, the equation

$$\cos x \left(3 \tan x + \frac{2}{\tan x} \right) = 4 \sin x - 5$$

giving your answers to 3 significant figures.

(4)**(Total for Question 10 is 8 marks)**

Q28**PAST PAPER**

Pearson IAL WMA12 · October 2025 · Q5

P2 · TOPIC 10**9 marks**

5. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < \theta \leq 360^\circ$, the equation

$$4 \tan \theta + 5 \sin \theta = 0$$

giving any non-exact answers to one decimal place.

(5)

- (ii) Solve, for $0 < x < \pi$, the equation

$$\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = \frac{5}{\cos x}$$

giving the answers, in radians, to 3 significant figures.

(4)

(Total for Question 5 is 9 marks)

03 Trig Models and Graphs

In $y = A \sin(kt + c) + D$: the maximum is $D + A$, the minimum is $D - A$, and the period is $\frac{360^\circ}{k}$ (or $\frac{2\pi}{k}$). To find “when”, set y equal to the value and solve like Idea 1 — then pick the time the question asks for.

WHEN YOU SEE → YOU DO

“the y coordinate of P (a maximum) is 7”

$$\longrightarrow D + A = 7$$

“the maximum temperature”

$$\longrightarrow \sin = 1: D + A$$

“the time of day when this occurs”

$$\longrightarrow kt + c = 90^\circ, \text{ then convert hours to a clock time}$$

“the first time after midday”

$$\longrightarrow \text{solve, then choose the first } t > 12$$

“how long above 0.5 m?”

$$\longrightarrow \text{solve for the two times in one period, subtract, count periods}$$

WORKED MODEL · JUNE 2024

$H = 10 + 12 \sin(kt + 18)^\circ$, $0 < k < 10$, $k = 6.41$. Maximum temperature and when.

1 $H_{\max} = 10 + 12 = 22^\circ\text{C}$

$$\sin = 1$$

2 $6.41t + 18 = 90 \Rightarrow t = 11.23 \text{ hours}$

$$\text{solve}$$

ANSWER 22°C at about 11:14 am

IDEA 03

Practise: Trig Models and Graphs

My questions first, then selected practice, then real past papers.

Q29**DR ESLAM**

A tide model

P2 · TOPIC 10

4 marks

$D = 6 + 2 \sin(30t)^\circ$, t hours after midnight. Find the greatest depth and the first time it occurs.

SELECTED PRACTICE

Q30**PRACTICE**

Practice · Medium

P2 · TOPIC 10

6 marks

- 9 (a) A seagull sits on the surface of the sea and moves up and down as waves pass.

Its height, h metres, above its position in calm water is modelled by the function

$$h = \frac{1}{2} \sin(180t) \text{ where } t \text{ is the time in seconds after timing commences.}$$

Sketch a graph of h against t for $0 \leq t \leq 10$ showing the coordinates of the points of intersection with the t - axis

(2 marks)

- (b) How many times in the first minute after timing commences is the seagull 0.25 metres above its calm water position?

(1 mark)

- (c) Find the time at which the seagull is first 0.25 m above its calm water position **and moving downwards**. Give your answer to 3 significant figures.

(3 marks)

Q31**PRACTICE**

Practice · Hard

P2 · TOPIC 10 6 marks**9 (a)**

(a) A seagull sits on the surface of the sea and moves up and down as waves pass.

Its height, h metres, above its position in calm water is modelled by the function

$$h = \frac{2}{5} \sin(180t)^\circ \text{ where } t \text{ is the time in seconds after timing commenced.}$$

Find the first time the seagull is 0.3 metres above its calm water position.

Give your answer to 2 decimal places.

(4 marks)**(b)**

(b) How many times in the first minute after timing commences is the seagull 0.3 metres above its calm water position?

(2 marks)

Q32**PRACTICE**

Practice · Very Hard

P2 · TOPIC 10**7 marks**

- 9 A seagull sits on the surface of the sea and moves up and down as waves pass.

Its height, h metres, above its position in calm water is modelled by the function

$$h = \frac{3}{5} \sin(90t)^\circ \quad \text{where } t \text{ is the time in seconds after timing commences.}$$

Find the amount of time the seagull is more than 0.5 metres above its calm water position in the first 20 seconds after timing commences.

Give your answer correct to 3 significant figures.

(7 marks)

PAST PAPERS

Q33

PAST PAPER

Pearson IAL WMA12 · January 2024 · Q9

P2 · TOPIC 10

14 marks

9.

In this question you must show detailed reasoning.

Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for
- $0 \leq x < 360^\circ$
- , the equation

$$\sin x \tan x = 5$$

giving your answers to one decimal place.

(6)

(ii)

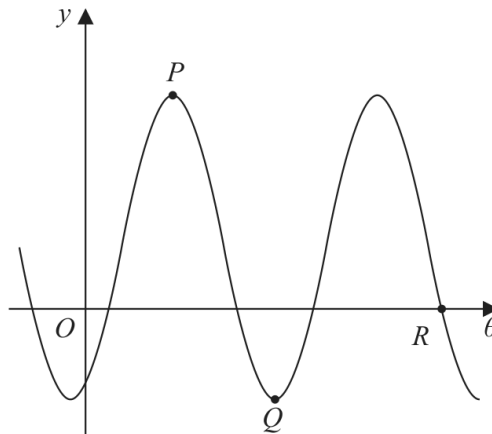


Figure 1

Figure 1 shows a sketch of part of the curve with equation

$$y = A \sin \left(2\theta - \frac{3\pi}{8} \right) + 2$$

where A is a constant and θ is measured in radians.The points P , Q and R lie on the curve and are shown in Figure 1.Given that the y coordinate of P is 7

- (a) state the value of
- A
- ,

(1)

- (b) find the exact coordinates of
- Q
- ,

(3)

- (c) find the value of
- θ
- at
- R
- , giving your answer to 3 significant figures.

(4)

(Total for Question 9 is 14 marks)

Q34 PAST PAPER Pearson IAL WMA12 · June 2024 · Q8

P2 · TOPIC 10 12 marks

8. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 < x \leq \pi$, the equation

$$5 \sin x \tan x + 13 = \cos x$$

giving your answer in radians to 3 significant figures.

(5)

- (ii) The temperature inside a greenhouse is monitored on one particular day.

The temperature, $H^\circ\text{C}$, inside the greenhouse, t hours after midnight, is modelled by the equation

$$H = 10 + 12 \sin(kt + 18)^\circ \quad 0 \leq t < 24$$

where k is a constant.

Use the equation of the model to answer parts (a) to (c).

Given that

- the temperature inside the greenhouse was 20°C at 6 am
- $0 < k < 20$

- (a) find all possible values for k , giving each answer to 2 decimal places.

(4)

Given further that $0 < k < 10$

- (b) find the maximum temperature inside the greenhouse,

(1)

- (c) find the time of day at which this maximum temperature occurs.

Give your answer to the nearest minute.

(2)

(Total for Question 8 is 12 marks)

Q35**PAST PAPER**

Pearson IAL WMA12 · June (R paper) 2024 · Q5

P2 · TOPIC 10**6 marks****5.**

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The depth of water, D metres, in a harbour on a particular day is given by the equation

$$D = 8 + 5 \sin\left(\frac{\pi t}{6} + 3\right) \quad 0 \leq t < 24$$

where t is the number of hours after **midnight**.

(a) Show that the depth of water in the harbour at 2 am is just over 4 metres.

(1)

(b) Find, to the nearest minute, the first time after **midday** when the depth of water in the harbour is exactly 6 metres.

(5)

(Total for Question 5 is 6 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- 01** $\sin(ax + b) = k$ New range for X ; CAST; undo $ax + b$.
- 02** **Identities** One function only; factorise; reject $|\cdot| > 1$.
- 03** **Models** Max $D + A$, min $D - A$, period $\frac{360}{k}$.

BOTTOM LINE One function → new range → every value with CAST → back to x . Never divide by $\sin x$.

Answers

Trigonometric Identities and Equations · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · SOLVE $\sin(ax + b) = k$: NEW RANGE, THEN CAST

- Q1** Dr Eslam · Transformed angle
 $X = 60, 300, 420, 660$: $x = 45^\circ, 165^\circ, 225^\circ, 345^\circ$
 $\tan \theta = 3$: $\theta = 71.6^\circ, 251.6^\circ$
- Q2** Dr Eslam · Sine equals k co-sine
- Q3** Practice · Easy
 (a) 5 cm (b)(i) $\frac{5}{13}$ (ii) $\frac{12}{13}$ (iii) $\frac{5}{12}$
- Q4** Practice · Medium
 third side $\sqrt{39}$; smallest angle opposite 5:
 $\sin \theta = \frac{5}{8}$, $\cos \theta = \frac{\sqrt{39}}{8}$, $\tan \theta = \frac{5}{\sqrt{39}} = \frac{5\sqrt{39}}{39}$
- Q5** Practice · Very Hard
 $\sin x = \frac{8}{\sqrt{89}}$, $\cos x = -\frac{5}{\sqrt{89}}$, $\tan x = -\frac{8}{5}$ (x is obtuse)
- Q6** Pearson IAL WMA12 · October 2019 · Q9
 (i) $\theta = 14.7^\circ, 85.3^\circ$. (ii)(a) Proven.
 (ii)(b) $\alpha = 5.517$ radians
- Q7** Pearson IAL WMA12 · October 2021 · Q10
 (i) $x = -\frac{11\pi}{24}, -\frac{7\pi}{24}, \frac{\pi}{24}, \frac{5\pi}{24}$. (ii) $\theta = 53.1^\circ, 180^\circ, 233.1^\circ$
- Q8** Pearson IAL WMA12 · January 2022 · Q7
 (i) $x = -73.3^\circ, 16.7^\circ$. (ii) $\theta = 1.69, 4.59$
- Q9** Pearson IAL WMA12 · January 2023 · Q8
 (i) $x = -0.94, -0.17, 1.15, 1.92$. (ii) $\theta = 70.5^\circ, 289.5^\circ$
- Q10** Pearson IAL WMA12 · June 2025 · Q8
 (i) $x = \frac{29\pi}{60}, \frac{59\pi}{60}$. (ii) $\theta = 50.3^\circ, 309.7^\circ$
- Q11** Pearson IAL WMA12 · January 2026 · Q8
 (i) $\theta = 51.9^\circ, 68.1^\circ, 141.9^\circ, 158.1^\circ$. (ii) $x = \frac{2\pi}{3}, \frac{4\pi}{3}$
- Q12** Pearson IAL WMA12 · June 2026 · Q2
 (i) $\cos^2 x - \cos x - 2 = 0$: $\cos x = -1$: $x = \pi$ (ii)
 $\tan(\theta + 60^\circ) = 4$: $\theta = 16.0^\circ$

IDEA 02 · THE TWO IDENTITIES: MAKE ONE TRIG FUNCTION

- Q13** Dr Eslam · One function only $2 \cos^2 x + 3 \cos x - 2 = 0: \cos x = \frac{1}{2}: x = 60^\circ, 300^\circ$
- Q14** Dr Eslam · Never divide by $\sin$ $\sin x(\cos x - \frac{1}{2}) = 0: x = 0^\circ, 60^\circ, 180^\circ$
- Q15** Practice · Easy $\frac{\sin^2 x}{\sin^2 x / \cos^2 x} = \cos^2 x$
- Q16** Practice · Hard $2 \cos^2 \theta + \cos \theta - 1 = 0: \cos \theta = \frac{1}{2} \text{ or } -1: \theta = -180^\circ, -60^\circ, 60^\circ, 180^\circ$
- Q17** Practice · Very Hard (a) $\cos 2\theta = 0 \text{ or } \sin 2\theta = \frac{1}{2}: \theta = \frac{\pi}{12}, \frac{\pi}{4}, \frac{5\pi}{12}, \frac{3\pi}{4}, \frac{13\pi}{12}, \frac{5\pi}{4}, \frac{17\pi}{12}, \frac{7\pi}{4}$ (b) $\cos 4x = \frac{2}{3}: x = 0.210, 1.36, 1.78, 2.93$
- Q18** Pearson IAL WMA12 · January 2020 · Q7 (a) Proven. (b) $x = 9.74^\circ, 80.26^\circ$
- Q19** Pearson IAL WMA12 · January 2021 · Q6 (a) Proven. (b) $x = 0, 0.365$
- Q20** Pearson IAL WMA12 · June 2021 · Q8 (i) $\theta = 36.8^\circ, 216.8^\circ$. (ii) (a) $a = 3, b = -7, c = 2$. (ii) (b) $x = -\cos^{-1}\left(\frac{1}{3}\right), 0, \cos^{-1}\left(\frac{1}{3}\right)$
- Q21** Pearson IAL WMA12 · June 2022 · Q5 $\theta = -13^\circ, 107^\circ$
- Q22** Pearson IAL WMA12 · October 2022 · Q5 (a) Proven. (b) $x = -1.13, -0.438, 0.225, 1.35$ radians
- Q23** Pearson IAL WMA12 · June 2023 · Q9 (a) Proven. (b) $x = 24.1^\circ, 155.9^\circ, 204.1^\circ, 335.9^\circ$
- Q24** Pearson IAL WMA12 · October 2023 · Q3 (a) $\theta = 131.8^\circ, 180^\circ, 228.2^\circ, 360^\circ$. (b) $x = 45.9^\circ$
- Q25** Pearson IAL WMA12 · June (R paper) 2024 · Q7 (i) $x = 1.318, 4.965$. (ii) $2\sqrt{2}$
- Q26** Pearson IAL WMA12 · October 2024 · Q9 (a) Proven. (b) $x = -2.50, -0.121, 0.645, 3.02$
- Q27** Pearson IAL WMA12 · January 2025 · Q10 (a) Proven. (b) $x = 3.48, 5.94$
- Q28** Pearson IAL WMA12 · October 2025 · Q5 (i) $\theta = 143.1^\circ, 180^\circ, 216.9^\circ, 360^\circ$. (ii) $x = 0.201, 2.94$

IDEA 03 · TRIG MODELS AND GRAPHS

- Q29** Dr Eslam · A tide model 8 m, when $30t = 90$: $t = 3$ (3 am)
- Q30** Practice · Medium (a) period 2 s, crosses at $t = 0, 1, 2, \dots, 10$ (b) 60 times (c) $180t = 150$: $t = 0.833$ s
- Q31** Practice · Hard (a) $\sin(180t) = 0.75$: $t = 0.27$ s (b) 60 times
- Q32** Practice · Very Hard $\sin(90t) > \frac{5}{6}$ for 0.746 s in each 4 s period:
 $5 \times 0.746 = 3.73$ s
- Q33** Pearson IAL WMA12 · January 2024 · Q9 (i) $x = 78.9^\circ, 281.1^\circ$. (ii)(a) $A = 5$. (ii)(b) $Q = \left(\frac{15\pi}{16}, -3\right)$. (ii)(c) $\theta = 5.51$
- Q34** Pearson IAL WMA12 · June 2024 · Q8 (i) $x = 1.91$ radians. (ii)(a) $k = 6.41$ or 17.59. (ii)(b) 22°C . (ii)(c) 11:14 am
- Q35** Pearson IAL WMA12 · June (R paper) 2024 · Q5 (a) $D(2) = 4.06 \dots$ m, just over 4 m. (b) 13:03, or 1:03 pm